

JEE MAIN REVISION MASTER: NUCLEAR CHEMISTRY

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 1 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Standard	Radioactivity, Nuclear Fission & Fusion	1 PYQs	Nuclear Chemistry

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Radioactivity, Nuclear Fission & Fusion	$t_{1/2} = \frac{0.693}{\lambda}$, $\text{Activity } A = \lambda N = -\frac{dN}{dt}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Radioactivity, Nuclear Fission & Fusion (Standard Yield — 1 PYQs)

Core Principle: Radioactive decay modes: α -emission lowers mass number by 4 and atomic number by 2; β^- -emission increases atomic number by 1 at constant mass number. Nuclear fission: ^{235}U captures thermal neutron, splits into lighter nuclei with release of ≈ 200 MeV and 2-3 neutrons. Nuclear fusion: lighter hydrogen nuclei combine at extreme temperatures to form helium with huge energy release.

■ **Caution / JEE Pitfall:** In β^- decay, a neutron converts to a proton and an electron inside the nucleus: $n \rightarrow p + e^- + \bar{\nu}_e$; the emitted electron does NOT come from atomic orbital shells!

■ **Speed Trick:** Heavy water (D_2O) and high-purity graphite are used as moderators to slow down fast neutrons in fission reactors.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "The synthetically produced radioactive noble gas by the collision of $^{98}_{249}\text{Cf}$ with $^{20}_{48}\text{Ca}$ is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#1)

The synthetically produced radioactive noble gas by the collision of ${}_{98}^{249}\text{Cf}$ with ${}_{20}^{48}\text{Ca}$ is

- (A) radon
- (B) radium
- (C) oganesson
- (D) xenon

Correct Answer: None

Solution Analysis: The reaction of ${}_{98}^{249}\text{Cf}$ and ${}_{20}^{40}\text{Ca}$ takes place as follows ${}_{98}^{249}\text{Cf} + {}_{20}^{40}\text{Ca} \rightarrow {}_{118}^{294}\text{Og} + 3{}_0^1\text{n}$. Thus, the radioactive noble gas formed is Oganesson ($Z=118$)